Lecture Notes · Preliminaries · Session 1 — The common language: sets, maps and proof

> The course's toolbox, in full rigor: sets and Cartesian products, maps (with the principle "finite: injective = surjective = bijective" proved), equivalence relations and partitions, logic and the four proof methods with complete examples. Exercises and a Going further section at the end. **First material of the course: nothing is assumed known.**
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> The course-wide conventions (★ priority exercises, optional **Going further**, interactive figures) live in How to use these notes, linked from the front page.

1.1 Sets

A **set** is a collection of objects (its **elements**). Notation: $x \in A$ (belongs to), $x \notin A$; by listing $\{1, 2, 3\}$ or in set-builder form $\{x : P(x)\}$ ("the x satisfying P"). The **empty set** $\emptyset$ has no elements.

Definition 0.1.1. $A \subseteq B$ (**subset**) if every element of A is in B. Two sets are **equal** if $A \subseteq B$ and $B \subseteq A$ — the **double inclusion**, which is *the* standard method for proving set equalities.

Operations: union $A \cup B = \{x : x \in A \text{ or } x \in B\}$; intersection $A \cap B = \{x : x \in A \text{ and } x \in B\}$; complement A^c (with respect to a universe); difference $A \setminus B$.

Proposition 0.1.2 (De Morgan's laws). $(A \cup B)^c = A^c \cap B^c$ and $(A \cap B)^c = A^c \cup B^c$.

Proof (double inclusion, first law). $x \in (A \cup B)^c \iff x \notin A \cup B \iff (x \notin A \text{ and } x \notin B) \iff x \in A^c \cap B^c$. The central equivalence is the negation of an "or" (§3.1). ■

> **Venn diagrams** make these laws evident at a glance — they are the intuition; the double inclusion is the proof. (In class, Venn; here, the proof.)

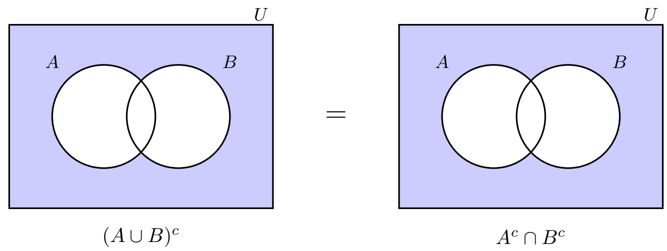


Figure 1: De Morgan's laws in Venn diagrams: $(A \cup B)^c = A^c \cap B^c$ — the outside of the union is the intersection of the complements.

1.2 Cartesian product

Definition 0.1.3. The **Cartesian product** $A \times B = \{(a, b) : a \in A,\ b \in B\}$ is the set of **ordered pairs** (where $(a, b) = (a', b') \iff a = a'$ and $b = b'$ — order matters, unlike in $\{a, b\}$). Iterating, $A^n = A \times \cdots \times A$ (n times): the **n-tuples** $(a_1, \ldots, a_n)$.

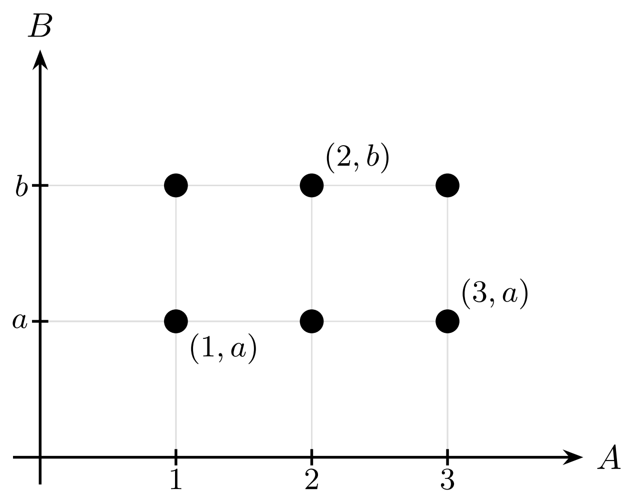


Figure 2: The Cartesian product $A \times B$ as a grid: each ordered pair (x, y) is a point; order matters.

> **Where this is going (announcement, not a dependency):** n-tuples will be the solutions of linear systems and the vectors of K^n (Arithmetic, Linear Algebra), and the complex numbers will be **defined** as pairs of reals (Complex Numbers). This humble definition

carries half the course.

1.3 Maps

Definition 0.1.4. A **map** $f : A \to B$ assigns to each $a \in A$ a unique $f(a) \in B$. A is the **domain**, B the **codomain**, and $f(A) = \{f(a) : a \in A\}$ the **image**.

Definition 0.1.5. f is:

- **injective** if $f(a) = f(a') \Rightarrow a = a'$ (it never sends two elements to one);
- **surjective** if $f(A) = B$ (it reaches everything);
- **bijective** if it is both (then the **inverse** $f^{-1} : B \to A$ exists).

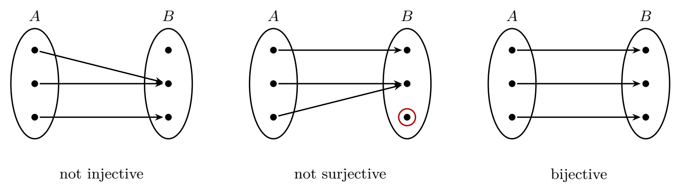


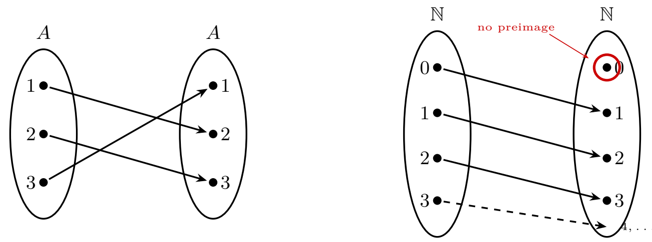
Figure 3: Injective, surjective, bijective in arrow diagrams: injective never sends two elements to one; surjective reaches all of B; bijective, both.

The **composition** $g \circ f : A \to C$ (with $g : B \to C$) is $(g \circ f)(a) = g(f(a))$; it is **associative**.

Theorem 0.1.6 (finite sets). If A is finite and $f : A \to A$, the following are equivalent: (i) f injective; (ii) f surjective; (iii) f bijective.

Proof. It suffices to show (i) $\iff$ (ii) (and then (iii) is both). Let $|A| = n$. The image $f(A)$ has exactly as many elements as distinct values taken by f. If f is **injective**, it takes n distinct values, so $|f(A)| = n = |A|$ and, since $f(A) \subseteq A$ with the same finite cardinality, $f(A) = A$: **surjective**. Conversely, if f is **surjective**, $|f(A)| = n$; but $f(A)$ has at most as many elements as distinct values, and if f were not injective it would take fewer than n values, giving $|f(A)| < n$ — contradiction. Hence injective. ∎

> **Announcement:** this principle — “on a finite set, injective already implies bijective” — is exactly the engine of the proof of **Fermat’s little theorem** (Arithmetic): multiplying by an invertible class is injective, hence a rearrangement. And the idea of a **map that respects structure** will be that of a **morphism** (Linear Algebra); the composition here, the matrix product there. (Theorem 0.1.6 **fails** on infinite sets: $n \mapsto n + 1$ on $\mathbb{N}$ is injective and not surjective — exercise 6.)



finite: inj. ⇒ bij.

ℕ: inj. not surj.

Figure 4: On a finite set, injective implies bijective (it is a permutation); on $\mathbb{N}$, $n \mapsto n+1$ is injective but not surjective (0 has no preimage).

1.4 Equivalence relations

Definition 0.1.7. A **relation** $\sim$ on A is an **equivalence** relation if it is:

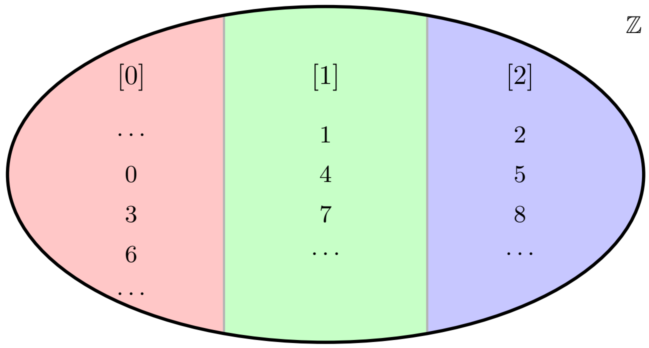
- **reflexive:** $a \sim a$;
- **symmetric:** $a \sim b \Rightarrow b \sim a$;
- **transitive:** $a \sim b$ and $b \sim c \Rightarrow a \sim c$.

The **class** of a is $[a] = \{b \in A : b \sim a\}$.

Theorem 0.1.8 (equivalence = partition). The classes of an equivalence relation form a **partition** of A: they are nonempty, their union is A, and two classes either coincide or are disjoint.

Proof. $a \in [a]$ (reflexive), so the classes are nonempty and cover A. If $[a] \cap [b] \neq \emptyset$, let c be a common element: $c \sim a$ and $c \sim b$, so $a \sim b$ (symmetric + transitive), and from there $[a] = [b]$ (everything equivalent to a is equivalent to b, by transitivity — double inclusion). ■

> **Announcement:** "$a \equiv b$: same remainder when dividing by n" will be an equivalence relation whose classes **are** the elements of $\mathbb{Z}/n\mathbb{Z}$ (Arithmetic). Theorem 0.1.8 guarantees that these classes split $\mathbb{Z}$ cleanly — the foundation of all modular arithmetic.



$\mathbb{Z}$ partitioned by $\equiv \bmod 3$

Figure 5: An equivalence relation splits the set into disjoint classes that cover it — example: congruence modulo 3 on $\mathbb{Z}$.

1.5 Logic and proof methods

1.5.1 Connectives, quantifiers and negation

Connectives: $\neg$ (not), $\wedge$ (and), $\vee$ (inclusive or), $\implies$ (implies), $\iff$ (if and only if). Quantifiers: $\forall$ (for all), $\exists$ (there exists). The key equivalences for proving:

- **Contrapositive:** $(P \Rightarrow Q) \equiv (\neg Q \Rightarrow \neg P)$.
- **Negation of connectives:** $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$; $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$ (logical De Morgan).
- **Negation of quantifiers** (a source of errors): $\neg(\forall x\, P(x)) \equiv \exists x\, \neg P(x)$; $\neg(\exists x\, P(x)) \equiv \forall x\, \neg P(x)$.

The **law of excluded middle** ($P \vee \neg P$ always true) is the logical foundation of proof by contradiction.

1.5.2 The four methods, each with a complete example

Direct ($P \Rightarrow Q$: assume P, deduce Q).

> **Example.** The sum of two even numbers is even. If $a = 2m$, $b = 2n$, then $a + b = 2(m + n)$. ■

Contrapositive (prove $\neg Q \Rightarrow \neg P$).

> **Example.** If n^2 is even, then n is even. Contrapositive: if n is odd, n^2 is odd. $n = 2k+1 \Rightarrow n^2 = 2(2k^2+2k)+1$, odd. ■ *(Doing it directly would be more convoluted — the contrapositive picks the easy path.)*

Contradiction (assume $\neg$(thesis) and reach an absurdity; uses the excluded middle).

> **Example.** *The canonical one: $\sqrt{2}$ is irrational.* If we had $\sqrt{2} = a/b$ with the fraction in lowest terms, then $a^2 = 2b^2$, so a^2 is even, so a is even (previous result), $a = 2c$; substituting, $b^2 = 2c^2$, so b is also even — but then a/b was not in lowest terms. Absurd. ■

Induction (on $\mathbb{N}$: prove the **base case** $P(1)$ and the **step** $P(k) \Rightarrow P(k+1)$).

> **Example.** $1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}$. Base $n = 1$: $1 = 1 \cdot 2/2$ ✓. Step: adding $k+1$ to the hypothesis, $\frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}$ ✓. ■
>
> *Variant:* **strong induction** (assume $P(j)$ for all $j \le k$) — the one Arithmetic will use for factorization.

Proposition 0.1.9 (well-ordering principle). Every nonempty subset of $\mathbb{N}$ has a **least** element. It is **equivalent** to the principle of induction.

> This is no triviality, but a property of the integers: in $\mathbb{R}$ it **fails** — the positive reals have no least element.
>
> **Announcement:** induction is the existence tool of the whole course (Euclidean division, unique factorization, De Moivre, dimension of vector spaces…), and its being "well founded" rests on exactly Proposition 0.1.9 — which Arithmetic will use explicitly, and by name, three times in its first session.

Remark 0.1.10 (a mention, not a section — enrichment). Proof by contradiction uses the excluded middle: from "it is not irrational" we conclude "it is rational". There is a school of thought, **constructivism** (or intuitionism, Brouwer), which **rejects** the excluded middle for infinite sets and, with it, certain pure proofs by contradiction — demanding, instead, that one *construct* the object. It is no mere curiosity: it underpins theoretical computer science (a constructive proof *is* an algorithm). A juicy nod for the Philosophy profile; not assessed material. *(More → Going further.)*

Exercises

1. ★ Prove "if n^2 is a multiple of 3, then n is a multiple of 3" in **two ways** (contrapositive and contradiction), and say which one you find cleaner.
2. ★ Negate correctly, minding the quantifiers: (a) "$\forall x\, \exists y : y > x$"; (b) "every student passed some exam"; (c) "$\exists x\, \forall y : xy = y$".
3. ★ For $f, g : A \to A$: prove that if $g \circ f$ is injective, then f is injective. Can anything be concluded about g? (Give a counterexample if not.)
4. ★ Prove by induction: $1 + 3 + 5 + \cdots + (2n-1) = n^2$, and identify exactly where you use the induction hypothesis.
5. Prove the second De Morgan law by double inclusion and draw its Venn diagram alongside.
6. Give a map $\mathbb{N} \to \mathbb{N}$ that is injective but not surjective, and another that is surjective but not injective: show that Theorem 0.1.6 **fails** on infinite sets.

7. Verify that "$a \sim b \iff a - b$ is even" is an equivalence relation on $\mathbb{Z}$ and describe its classes (how many are there?). *(It is the case $n = 2$ of what in Arithmetic will be $\mathbb{Z}/n\mathbb{Z}$.)*
8. Prove that the composition of two injective maps is injective, and the composition of two surjective maps is surjective.
9. **(Excluded middle in action.)** In the proof that $\sqrt{2}$ is irrational, point out the exact spot where "rational or irrational, there is no third option" is used, and reflect on why a constructivist would ask for something more.

Going further and references

Going further (cardinality: some infinities are bigger than others). Two sets have the "same size" if there is a bijection between them — and with this definition, $\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ have **the same** size (they are *countable*), but $\mathbb{R}$ is **strictly larger** (Cantor's diagonal argument). Infinity comes in sizes. An enjoyable introduction: Halmos, *Naive Set Theory*.

Going further (constructivism and computation). The intuitionistic rejection of the excluded middle, far from being a curiosity, is the basis of the **Curry–Howard correspondence**: "proofs = programs". In modern proof assistants (Lean, Coq), a constructive proof of "there exists x" *contains* an algorithm that produces it. Philosophy of mathematics with very concrete consequences — natural territory for the joint degree in Mathematics and Philosophy.

References. Logic and proof methods: Velleman, *How to Prove It* (excellent and elementary); any "discrete mathematics" text (Biggs). Sets and maps: Halmos, *Naive Set Theory*; Dorronsoro–Hernández, ch. 1.